

SHIV JHALANI

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SKILLS

- **Languages & Technologies:** Python, C++, SQL, Java, AWS (Lambda, Step Functions, Athena, EC2, etc.), Flask
- **Areas of Focus:** Machine Learning, Agentic Workflows, Natural Language Processing, API Design & Development

WORK EXPERIENCE

INRIX, Backend Software Engineer August 2025 - Present

- Reduced customer-facing data delivery lag by **57%** by migrating critical data pipelines for Trips product
- Modernized 200+ production **AWS Lambdas**, upgrading legacy **Python** runtimes & standardizing deployments through CI/CD
- Designed database restructure to reduce monthly storage by 90%, saving space on over 100M+ rows by removing redundant data and preventing cartesian explosion, improving read/write efficiency by 25%, and saving over **\$20K** in costs annually
- Automated provider eval workflow, cutting manual effort by 80%, saving 20+ developer hours/month, while improving reliability
- Proposed & led a geospatial ML initiative from idea to production, now [used across multiple customer-facing products](#)
- Authored & published a [blog post](#) for INRIX's official blog series, highlighting the use of geospatial data in mobility solutions
- Designed & implemented multiple end-to-end backend pipelines for new mobility products using **Python, Java & AWS Services**
- Led 6-person **AI-native backend effort** with Athena and ClickHouse, enabling synchronous retrieval from over **10TB+** of data
- Led AI workshops and GenAI hackathons at Santa Clara University, engaging INRIX & AWS leadership with 300+ participants

INRIX, Backend Software Engineering Intern June 2024 - September 2024

- Enhanced efficiency of [customer-facing](#) API using **Python, Java, AWS Athena & AWS Lambda**, cutting costs up to **90%**
- Upgraded legacy code to improve performance & maintainability while ensuring backward compatibility

INRIX, Software Engineering Intern June 2023 - September 2023

- Implemented automated multithreaded data ingestion pipeline for an internal tool using **Java, Python, & AWS Lambda**
- Resolved critical error-prone data update process, improving efficiency by **20%**, deployed code to production & tested via CI/CD
- Led 11-member team to final round of company-wide hackathon, through global cross-team collaboration & project management

EDUCATION

Santa Clara University, College of Arts & Sciences (Santa Clara, California) September 2021 - June 2025
Bachelor of Science in Computer Science (Data Science Emphasis), Minor in Mathematics GPA: 3.96

- **Relevant Coursework:** Data Science, Data Mining, Applied Machine Learning, Discrete Mathematics, Linear Algebra
- **Activities & Societies:** Theta Tau Professional Engineering Fraternity, ACM, Career Center, Center for Entrepreneurship
- **Awards and Honors:** Alpha Sigma Nu, Phi Beta Kappa, Upsilon Pi Epsilon, Student Life Award, University Honors Program

RELEVANT RESEARCH

[Clickbait Detection on Social Media Platforms using Topic Modeling](#), *Research Assistant* January 2024 - June 2025

- Collaborated with team of 2 to detect clickbait on social media through **OpenAI APIs & topic modeling algorithms** like LDA
- Utilized OpenAI APIs to summarize text columns, transform into vectors using sentenceTransformers library & employ cosine similarity to **quantify correlation** between the title & transcripts on two different datasets of a total of over 7700 rows
- Co-authored research paper detailing significant findings, **presented & published** at the 2024 IISA conference

[Online Hate Speech Spreader Detection using Modified PageRank Algorithm](#), *Research Lead* October 2021 - January 2024

- Led team of 3 to detect & flag hate speech spreaders on online social networks using **Python** & libraries like sklearn
- Modified Google's PageRank algorithm to implement **graph-based** model using adjacency matrices to calculate user influence & measure hate speech spread within user networks, enabling **data & network analysis** on pre-trained **machine learning** models
- Co-authored research paper detailing significant findings, **presented & published** at the 2023 International DSML conference

RELEVANT PROJECTS

[Post Filtering Extension](#), *Backend Developer (1st Place Winner - Replit Award, xAI Hackathon)* October 2024

- Won **1st Place** out of 500+ participants for the Replit track, beating out over 80+ teams
- Designed & implemented backend architecture of Chrome extension that filters out engagement farmers, self-promoters, & bots on X using **Grok 2 LLM** and **xAI APIs**, fostering relevant and meaningful conversations
- Integrated X APIs to retrieve and process posts in a streaming architecture, enabling incremental data delivery & reduced latency

[Agentic Harness Container Builder \(VAEN\)](#), *Creator/Open-Source Developer* April 2026

- Built **open-source Python CLI** that packages **agent instructions**, skills, **MCP** server definitions, and client-specific config into portable **.agent** bundles, supporting agentic coding tools like **OpenAI Codex**, **Claude Code**, **GitHub Copilot**
- Designed a manifest-driven build system that serializes setups into deterministic OCI image-layout archive with a supporting import workflow that materializes archived setups into expected locations for agentic coding workflows

[Agentic Tool Registry](#), *Creator/Open-Source Developer* February 2026

- Designed & implemented **open-source Python CLI** metadata registry enabling **coding agents** to discover external tools, APIs, MCP servers, & CLIs through structured agent-ready tool records with a validation pipeline to verify tool metadata for security
- Engineered progressive lookup pipeline while prioritizing lightweight discovery metadata, **reducing agentic context overload**